

CONTINUATION

Helix OTA — Continuation / Resume Handoff

Field	Value
Revision	5
Created	2026-06-07
Last modified	2026-06-11T13:15:00Z
Status	active — resume with “continue”
Status summary	Single source of truth for resuming work. Captures exactly what is DONE (verified), the git state, and the prioritized NEXT steps. Everything below is committed to main and pushed to all 4 upstreams (GitHub, GitLab, GitFlic, GitVerse).

↗ CURRENT STATE (2026-06-11) — HEAD 42be557 — read docs/RESUMPTION.md FIRST

PWU-AB milestones landed GREEN + pushed (HEAD 42be557, all 4 upstreams). On the emulator A/B ladder (T1 = QEMU virt + HVF on this macOS host, real U-Boot 2024.01): **PWU-AB-1 FULL A/B slot switch is PROVEN** (commit 18ed84a; evidence docs/qa/20260611T094958Z-ab-slot-switch/ — Run A→slot A /dev/vda2, Run B→slot B /dev/vda3, the slot genuinely switched, §11.4.50 determinism 2/2 identical per the captured determinism_soak.txt) **AND PWU-AB-3 corrupt-slot AUTO-ROLLBACK is PROVEN** (commit 42be557; evidence docs/qa/20260611T095918Z-ab-rollback/ — U-Boot bootcount=2 > bootlimit=1 → altbootcmd swap → fell back from bad slot B to good slot A, with a CONTROL run proving the rollback fires only on a bad slot). **PWU-AB-2 RAUC dm-verity** (tests/emulator/ab_virt/ab_rauc_verity.sh) is **AUTHORED** but **UNVERIFIED-pending** — needs a signed .raucb bundle + reconciling the RAUC slot-class scheme with the proven boot.cmd BOOT_ORDER env scheme. **T2 Cuttlefish** real-Android-A/B remains **SKIP-pending** the operator’s incoming Linux + nested-KVM host (/dev/kvm absent on this Apple-Silicon host); **T3 RK3588** hardware **PENDING** (no board). Status docs: docs/emulator/

rk3588_ab_virt/Status.md (Rev 3) + Status_Summary.md (Rev 3). **No release tag** — §11.4.40 needs the full ladder GREEN (T2 + T3 still SKIP/PENDING), so a tag would be a bluff (§11.4.6). Everything below this box is prior-wave history.

↗ **PRIOR STATE (2026-06-10 overnight) — superseded by the box above**

The freshest live-state anchors are in **docs/RESUMPTION.md** (the §11.4.131 canonical entry). As of HEAD **0d27438** the build is **GREEN across every achievable tier**, and during the 2026-06-10 overnight autonomous session (operator away-from-keyboard) **every achievable software tier was RE-RUN fresh on this exact HEAD and re-confirmed GREEN** (§11.4.132 risk-ordered re-validation) — full proof in **docs/qa/STABILITY_REPORT.md** Rev 2: Go (all tiers + -race -count=1), pgx integration (store 88.7% / rollout 71.8% with -tags integration), inheritance gate, meta-test §1.1, dashboard 58/58 + typecheck, HelixQA LIVE bank 10/0 + dry-run 10/0 + self-test §1.1, podman full-lifecycle + fleet 5/5 + recall-recovery 7/7, QEMU firmware tier, all Go submodule cores. Android bricks (ota-android-agent 1061015, ota-update-engine-bridge 8bb8d2f) are byte-identical to their prior proof (pins unchanged, trees clean at pin) — NOT re-run, to respect the §12.6 60%-memory ceiling alongside podman (honest §11.4.6/§11.4.101 call, not a re-run claim). **No release tag** — §11.4.40 needs the on-device RK3588 tier (hardware-BLOCKED, §11.4.112), so a tag would be a bluff. **BLOCKED (hardware):** T2 Cuttlefish / GSI-A-B real-apply / T3 RK3588. **DEFERRED (zero-risk):** fabric scheduler P3 (unwired §11.4.124), HelixQA in-tree compile (./containers layout). **UPDATE 2026-06-11: an operator-directed 6-stream parallel-hardening round landed on top (HEAD dead6ef) — real coverage gains (api 87.7% / fabric 96.7% / transport 94.9%, dashboard 58→93), §11.4.65 PDF backfill (66 siblings — weasyprint IS present, the prior “no weasyprint” deferral was stale), and an extended security suite (26/0/0 ×3); two real defects in the agents’ output were caught + fixed during conductor-verification. A SECOND parallel round then landed (HEAD a58f7f8): the integration-port collision is now FIXED (flock-serialized shared Postgres), deviceemu 76→94%, ota-protocol + ota-artifact-validator submodules → 100% (pushed eda12b7/087fa08), a rollout halt-on-breach e2e (47/0/0), and the extended security suite wired into the HelixQA bank (LIVE 11/0/0). Remaining follow-ups are now DIMINISHING-RETURNS (other-submodule coverage, more e2e scenarios) — see docs/RESUMPTION.md §1. Everything below this box is prior-wave history.**

How to resume

Type **continue** in a new session. Read **docs/RESUMPTION.md Rev 8** + this file + the memory index first. All work is on main (latest commit pushed to all upstreams). Working branch history: foundation → research → MVP specs → submodule impls → server, each merged to main.

Locked decisions (do not re-litigate)

- Stack: **Go + Gin + Brotli + HTTP/3(QUIC)→HTTP/2 fallback, REST primary**; reuse vasic-digital/http3.
- Strategy: **native Android A/B (update_engine + AVB/dm-verity + auto-rollback) + custom Go control plane**; wrap OSS only where it adds value (hawkBit gated, ADR-0001).
- Trust: signing + SHA-256 + AVB for MVP; TUF server-side then device-side in 1.0.1 (ADR-0002).
- Topology: modular monolith for MVP (ADR-0003).
- New submodule repos auto-created PUBLIC on GitHub + GitLab (pre-authorized).
- docs/research/main_specs/additions/ files are authoritative input — always analyze + fold in.
- Commit + push to ALL upstreams regularly. Merge to main when a milestone is done.

Session 2026-06-08 update (DONE, on main, pushed all 4 upstreams)

- **Constitution submodule relocated** HelixConstitution/ → **constitution/** (repo/URL unchanged) via git mv; fixed the only filesystem path refs (tests/test_strategy.md, additions/initial_research_01.md) incl. their pre-existing off-by-one link depth. Prose “HelixConstitution” clause citations (the repo NAME) intentionally left.
- **Inheritance wired**: parent CLAUDE.md, AGENTS.md, docs/guides/HELIX_OTA_CONSTITUTION.md inherit from the submodule; tests/inheritance_gate.sh (5 invariants, matches the exact §11.4 forensic-anchor heading) + tests/test_constitution_inheritance.sh (gate green clean + **§1.1 paired-mutation proven** via constitution/meta_test_inheritance.sh + recursive submodule pointer check) + tests/pre_build_verification.sh + installed pre-commit hook. All 6 owned ota-* submodules carry CLAUDE.md+AGENTS.md inheritance pointers (pushed github+gitlab; gitlinks bumped to v0.1.0-1-g...). Commits ee5dc7d, e11a221.
- **All 3 additions processed** (operator mandate — nothing skipped): per-input exhaustive inventories in research/additions_analysis/0{1,2,3}_analysis.md; research/additions_synthesis.md → **Rev 2** with consolidated 14-gap register (§8), new-work routing (§9), addition-#3 conflict reconciliation (§10, locks Gin/ed25519/JWT/releases+deployments/≥90%), future-phase catalogue (§11), UNVERIFIED register (§12). MVP-critical **G1 (anti-downgrade invariant)** implemented as regression tests; handleClientUpdate 79.3%→86.2%. Commit 629b4eb.
- **Constitution path is now constitution/** — update any new references accordingly.

Server hardening — DONE this session (handoff NEXT #1, real evidence)

- **Coverage**: upload handler 71.2%→**96.6%**, client-update 79.3%→**93.1%** (≥90% safety floor). Commit 64c3bfd.

- **pgx PostgreSQL Repository** (server/internal/store/postgres.go): full store.Repository impl; shared contract_test.go proves parity with memory; **integration test boots real Postgres via the containers submodule on podman** (go test -tags integration ./internal/store/), evidence in docs/qa/20260608-pgx-postgres-integration/. Surfaced+fixed a real idempotency overwrite bug. Commit 96fdecb.
- **Brotli→gzip→identity** compression middleware (server/internal/api/compression.go). Commit b26f30e.
- **HTTP/3 (QUIC)+HTTP/2 fallback** via the new submodules/http3 (digital.vasic.http3) — server/internal/transport/; wired into cmd/ota-server (TLS via HELIX_TLS_CERT/KEY → HTTPS port 8443; plain HTTP otherwise). Real h3+h2 client test; evidence in docs/qa/20260608-http3-h2-brotli-transport/. Commit 1469edf.

Parallel-wave deliverables — DONE this session (commit 8efb6b8)

6 subagents closed additions gaps: **G2** validator hash-before-signature VERIFIED (FACT, file:line); **G7** staged-rollout engine spec + migration_002 design; **G6** dashboard design; **G3/G4/G5** operational endpoints spec (audit/telemetry-reads/group CRUD + proposed repo methods); **G10** CI (.github/workflows/ci.yml + CODEOWNERS + dependabot); **§11** future-phase folding (1.0.2-rollback, 1.0.3-delta-updates created; 1.X-linux/windows/other-os extended).

Operational endpoints — DONE this session (G3/G4/G5, real-Postgres-verified, memory+pgx parity)

- **G3 audit** (commit 9a703bf): auditMiddleware records successful mutating actions (reads/failures skipped, ids never leak into the action) + GET /audit (admin) + audit_logs table.
- **G4 telemetry reads** (commit eadaa7f): GET /devices/{id}/telemetry (device reads only its own) + GET /telemetry/overview (fleet counts).
- **G5 device-group CRUD** (commit e3e1307): full /groups CRUD + membership; writes operator/admin, delete admin-only. All three added store.Repository methods on memory + pgx, extended the shared contract, and pass the pgx integration test on real Postgres.

Staged rollout + more — DONE this session

- Migration 002 real-DB validated (e14942a). **Go engine** wired via the ota-rollout-engine brick: server/internal/rollout + REST POST/GET /deployments/{id}/rollout + /evaluate (create→start→advance→complete, halt-on-error-breach); 7 tests green (046ea07).
- End-user rollback reconciled INTO **1.0.1** per operator decision; 1.0.2-rollback superseded→folded (a4c2b3d).
- **G11** repo audit: all 6 ota-* PUBLIC on GitHub+GitLab (dab2f0e). **G12** NFR/load harness real measured percentiles (6547c7f).

Round 2 deliverables — DONE this session

- **Dashboard** scaffold (G6) — Vite+React+TS, real tsc/vite build exit 0 (813a00e).
- **6 submodule READMEs** enriched + pushed to each github+gitlab (813a00e).
- **Dependabot ×4 merged + verified:** gin 1.12.0, quic-go 0.60.0 (http3 transport re-verified), actions v6 (33039b8).
- **Rollout pgx StoragePort** — real-DB tested via containers submodule (8190e92).
- **§11.4.65 exports** for 13+2 new docs (242ee2f, a288a4b). **device_tuf.md** + **rollback_ux.md** specs (1.0.1).

Round 3 deliverables — DONE this session

- **Rollback-history store layer** (AppendRollback/ListRollbacks) memory+pgx, real-DB parity (e4165a6).
- **Recall endpoint** POST /deployments/{id}/recall + GET /rollbacks (c98dfac) — records rollback_history, validates deployment/target-release, operator/admin; 3 tests.
- **Delta-updates 1.0.3** design (ADR-0005 Option B) + **as-built endpoint reference** (server.go route table) + submodule README §11.4.65 siblings (573779a).
- **Autonomous e2e challenge** tests/e2e/challenge_operational.sh — real live-server run, 28 passed/0 failed/1 skip, independently re-verified (bb332a4).

Round 4 deliverables — DONE this session

- **pgx wired into main** (HELIX_DATABASE_URL): production uses the pgx repo + rollout StoragePort; **e2e challenge PASS (28/0/1) against a real Postgres-backed server** (1cdad81, docs/qa/20260608-pgx-server-e2e/).
- **migration 004** delta_artifacts real-DB validated; **CI** e2e+loadtest jobs; **README** doc-map §10; **HelixQA bank**; **threat-model** extended (6a3e213, c-threat).

Round 5 deliverables — DONE this session

- **Recall = forward-fix** (operator decision honor-AVB, 4e35c3e): handleRecall supersedes the current deployment + creates a NEW active deployment of the target release; the update-check anti-downgrade invariant means AVB is honored by construction. New store UpdateDeployment (memory+pgx, real-DB parity). Decision recorded in rollback_ux.md Rev 2 + threat_model.md §11.11 RESOLVED (f5ec504).

Round 6 deliverables — DONE this session

- **Delta-artifact store + API** (66464d7): delta_artifacts (base≠target CHECK + UNIQUE pair) on memory+pgx; POST/GET /deltas register+lookup; real-DB parity.
- **device-TUF client-decision memo** (recommend gomobile-go-tuf/v2, ADR-0002 §4.3) + sibling.

- **Additive WIDENs — full set landed** (operator-approved): audit ?since/? until filters + telemetry failure_rate/by_state (028e656, memory+pgx real-DB parity) + group member_count (4cb86d7). spec_impl_alignment.md Rev 3: rows 2+5+6-partial landed; as-built doc re-synced (recall now WIRED, / deltas + recall + rollbacks documented).

Round 7 — breaking WIDENs landed (operator ruling: WIDEN-impl)

- **Group** (a91271b + 02ad2d0): id→group_id; batch member-add → 200 {added, already_member, not_found}. Server+tests+e2e(36/0/1)+dashboard+bank+docs.
- **Audit** (fbaefbe): actor → object {user_id, subject}. Server+test+dashboard+docs.
- **Telemetry per-device** (2a48ab5): events→items, newest-first, ?limit/? cursor+next_cursor. Server+tests+dashboard+docs.
- **GET /members** (3b4b1d8 + e2e): device_ids[] → items[] of {device_id, added_at}; store gained device_group_members.added_at + ListGroupMembersDetailed (memory+pgx, real-DB parity); e2e 39/0/1.
- Each: full default suite green, all wire consumers updated in lockstep, spec_impl_alignment.md rows 6/8/1/4-structural + GET-members RESOLVED.

WIDEN ruling status: COMPLETE except two legitimately-parked items: row-4 richer telemetry fields (duration_ms/bytes_transferred) **blocked on UNVERIFIED ingest** (event source must carry them first) + the per-device telemetry filters; group/members list pagination (row 7) **deferred** (groups bounded — memo’s own recommendation). Both need either ingest work or an operator nudge; not autonomously actionable now. *(UPDATE 2026-06-10: the per-device telemetry **filters** + group/members list **pagination** are now DONE — shipped 50ef5c6; see the “Session 2026-06-10 update” below. Only row-4 richer numeric fields remain parked on ingest.)*

Round 8 — “all of it” wave (operator: do all frontiers)

- **Server-side delta-selection** (9179d0a): ota-protocol UpdateAvailable.Delta (brick 7d18edc, pushed; server builds via dev replace); store ReleaseByVersion (memory+pgx, real-DB parity); update-check resolves current→base artifact→FindDelta → offers delta + full fallback; TestClientUpdateOffersDelta.
- **Device-side delta-apply** (ota-android-agent, pushed + gitlink): pure DeltaApplyDecision (USE_DELTA vs FULL_PAYLOAD), 11 tests, :core 47/0 real Gradle.
- **Operator cleanup** (DONE): vasic-digital/containers + HelixConstitution GitLab mirrors flipped **private**→**public** (re-read-verified); CODEOWNERS @milos85vasic confirmed.
- **OpenAPI** synced to all widened/new endpoints + UpdateAvailable.delta; §11.4.65 corpus export coverage (46 html+pdf pairs).

Round 9 — hardware-free QA wave (operator: “all of it”, no Orange Pi tonight)

- **Stress + chaos** (§11.4.85, 5363511): race-clean in-process suite — 200 concurrent creates (0 err, p99 6.4ms), 2400 sustained reads (0 err), 60-device membership contention → no lost updates, chaos fault→500→recover→200. Evidence docs/qa/20260608-stress-chaos/.
- **Full-pipeline e2e + security** (e7e1alc): tests/e2e/pipeline_signed.sh (real ed25519-signed artifact → upload→release→deploy→rollout→delta-bearing update; bogus-sig→422) 32/0/0 — **closes the artifact-pipeline SKIP**; tests/security/security_probes.sh (authn/authz/ownership/injection/trust-boundary) 37/0/0. Both re-run in-tree.
- **Performance/NFR + scaling** (7b75212): real memory-vs-pgx concurrency sweep — in-mem 20.5k RPS @ c=128 p99 30.6ms; pgx ~6.2k plateau (2-CPU container). docs/research/main_specs/1.0.0-mvp/nfr/performance_baseline.md.
- **Dashboard build-out + Playwright** (just landed): functional Fleet/Deployments+Recall/Groups/Audit screens on the real API; tsc+build exit 0; **Playwright 5/5** vs a live server.
- **Coverage ledger** (§11.4.25) + **HelixQA bank** now 8 challenges (incl. signed-pipeline + security).
- **Benchmarking** (§11.4.27, bbc97e4): Go testing.B suite — healthz 2.5µs, group-create 8.5µs, update-check 4.9µs, FindDelta 160ns/0-alloc, etc. docs/qa/20260608-benchmarks/.
- **DDoS/flood probe** (§11.4.27, 6d29fb8): 5,952-req burst served, responsive post-flood; surfaced the honest “no rate-limiter” finding.
- **Rate-limit FEATURE** (1b97fc7): implemented the finding’s fix — in-flight cap middleware (HELIX_MAX_INFLIGHT, default-off) sheds 429 RATE_LIMITED; proven cap=1/300-concurrent → 244 shed / 56 served / recovers. **§11.4.27 test-type matrix now complete** for the operational surface (unit/integration/e2e/security/ddos/scaling/chaos/stress/performance/benchmarking/ui/Challenges). Full rebuild+validate sweep green (docs/qa/20260608-full-rebuild/).

Session 2026-06-10 update (DONE, on main, pushed all 4 upstreams)

- **Per-device telemetry filters + group/members pagination** (50ef5c6, feat(api): per-device telemetry filters + group/members pagination): closes the two previously-PARKED WIDEN bits — per-device telemetry now accepts the filter params, and the group/members list is paginated. **OpenAPI synced + redocly-clean**. These move out of the “parked WIDEN bits” list (no longer deferred).
- **Dashboard client lockstep** (b0b8ee2, chore(dashboard): sync API client to new pagination/filter params): the dashboard API client updated in lockstep with the new pagination/filter params, so every wire consumer stays consistent (§11.4.92 cross-feature consistency). **HEAD is now b0b8ee2**.
- **Emulator-driven device testing initiative started** — tiered plan captured as FACT in docs/design/EMULATED_DEVICE_TESTING.md. **Tier-1 (podman ota-device-emulator over real ota-protocol) is IN PROGRESS**; Tier-2 (Cuttlefish A/B, Linux+nested-KVM-gated) and Tier-3 (real RK3588, hardware-gated) are designed, host/hardware-gated (NOT structurally

impossible per §11.4.112). Extends the containers submodule (pkg/boot/compose/health + pkg/emulator AVD-x86_64 + pkg/vm qemu-aarch64) per §11.4.76.

- **Canonical §11.4.131 session-resumption file created** at docs/RESUMPTION.md — the fixed out-of-the-box entry point for any fresh session (SHORT + FULL variants, read-first handoff pointers, live-state anchors, PHASE/NEXT/terminal-goal, binding constraints). Point a new session at that one file.

NEXT wave (PHASE: emulator A/B ladder — slot-switch + rollback DONE; dm-verity next, then hardware-gated tiers)

1. **PWU-AB-2 RAUC dm-verity (the immediate NEXT, software-doable on this host)** — run tests/emulator/ab_virt/ab_rauc_verity.sh: produce/sign a .raucb bundle and reconcile the RAUC slot-class scheme with the PROVEN uboot_ab/boot.cmd BOOT_ORDER env scheme, then capture dm-verity slot-integrity evidence (E4). Currently AUTHORED + UNVERIFIED-pending.
2. **Tier-2 Cuttlefish real-Android A/B** (tests/emulator/tier2_cuttlefish_ab.sh, incl. PWU-CF-2 corrupt-slot rollback) — SKIP-pending the operator's incoming **Linux + nested-KVM host** (/dev/kvm absent on this Apple-Silicon macOS host). Ready to run there.
3. **Tier-3 real RK3588 / Orange Pi 5 Max on-device e2e** (download→verify→A/B apply→reboot→post-boot; corrupt-slot→auto-rollback) — PENDING, no board on the bench.
4. **Device-side TUF** (gomobile-go-tuf/v2 per the decision memo) — gated on an arm64 .so-size/JNI measurement on real RK3588 hardware.
5. **Device payload-apply integration** — wire DeltaApplyDecision into the on-device apply path (:android/update_engine) — needs a real device to validate end-to-end.
6. **Parked WIDEN bits**: row-4 richer telemetry fields (duration_ms/bytes_transferred) — still blocked on UNVERIFIED ingest (event source must carry them first). Telemetry per-device filters + group/members list pagination are DONE (shipped 50ef5c6).

DONE this session (2026-06-11, on main, pushed all 4 upstreams): PWU-AB-1 FULL A/B slot switch (18ed84a, evidence docs/qa/20260611T094958Z-ab-slot-switch/, determinism 2/2 identical) + PWU-AB-3 corrupt-slot AUTO-ROLLBACK (42be557, evidence docs/qa/20260611T095918Z-ab-rollback/), both against real U-Boot 2024.01 + QEMU virt + HVF. **HEAD is now 42be557.**

Carried-forward gaps register

See additions_synthesis.md §8/§9 (14 gaps; most now specced — implementation pending). Numbering decision: 1.0.1 = staged-rollout; rollback→1.0.2, delta→1.0.3.

DONE (verified, on main, pushed)

1. **Spec corpus** (docs/research/main_specs/): master design + ADRs index; research = scored landscape report + 12 stack notes + 5 ADRs

- (ADR-0001..0005) + additions synthesis; foundation = glossary, STRIDE threat model, submodule reuse map, doc standards, requirements traceability, export pipeline; **1.0.0-mvp** full specs (api+OpenAPI, database+migrations, security, server, client_android+snippets, deployment, tests, VALIDATION_EVIDENCE); 6 Mermaid diagrams; future-phase outlines (1.0.1, 1.X-linux/windows/other-os); corpus README.
2. **Validation evidence (real)**: OpenAPI redocly-valid; **migrations applied to live PostgreSQL** (12 tables up, clean down); k8s kubeconform 5/5; compose YAML valid; all 6 diagrams render (mmdc); corpus exports = 50 HTML + 50 DOCX + SVG/PNG (scripts/export_docs.sh). PDF needs a LaTeX engine (not installed).
 3. **6 reusable submodules** (created on GitHub HelixDevelopment/ + GitLab helixdevelopment1/, scaffolded, wired under submodules/, **implemented v0.1.0, tagged, pushed both hosts**):
 - ota-protocol (Go, 99.3%), ota-telemetry-schema (Go, 98.9%), ota-artifact-validator (Go, 97.8%, ed25519+sha256), ota-rollout-engine (Go, 94.9%, halt-wins+deterministic cohorts)
 - ota-update-engine-bridge (Kotlin; :core 27 tests; :android builds an AAR)
 - ota-android-agent (Kotlin; :core 36 tests; :android build pending KMP/AGP-on-Gradle-9.5 alignment)
 4. **Submodules wired**: HelixConstitution + containers at root; 6 ota-* under submodules/.
 5. **Control-plane server** (server/, Go module github.com/HelixDevelopment/helix_ota/server): Gin modular monolith wiring the 4 Go modules; full /api/v1 MVP endpoints (auth, devices, artifacts upload+validate S1-S6, releases, deployments all-targets, client/update 204|200, telemetry, health); **66 httptest integration tests pass**; gofmt/vet/build clean.

NEXT (prioritized)

1. **Server hardening to spec**: raise internal/api coverage to $\geq 90\%$ on handleUploadArtifact + handleClientUpdate (error-path branches); add a **pgx** Repository implementation behind the existing store.Repository interface + testcontainers-go integration tests against real Postgres; wire vasic-digital/http3 for HTTP/3→HTTP/2 + Brotli.
2. **ota-android-agent :android build**: switch its :android module from Kotlin MPP to plain com.android.library + org.jetbrains.kotlin.android (like the bridge, which builds), OR pin Gradle 8.x + AGP 8.7.x via a wrapper; then produce the AAR.
3. **React dashboard** (dashboard/): reuse UI-Components-React, Dashboard-Analytics-React, Auth-Context-React; secure login, upload, deploy, fleet health.
4. **1.0.1 staged-rollout**: promote the outline to full specs + implement rollout API (phases, pause/resume/abort), device-side TUF, end-user rollback; migration 002_* (deployment_phases, rollouts, rollback_history, TUF metadata).
5. **AOSP integration**: build the agent + bridge as a system/priv-app in the Orange Pi 5 Max AOSP tree; on-device e2e (download→verify→A/B apply→reboot→post-boot; corrupt-slot→auto-rollback).
6. **Containerize** the dev stack via the containers submodule; CI running all validators (redocly, Postgres migrations, kubeconform, go test, gradle :core:test, export render).

7. **Per-submodule:** full README/docs/manuals + GitFlic/GitVerse mirrors + tag mirroring (§4).

Toolchain facts (this host)

Go 1.26.2; Android SDK (platforms 31–36.1, build-tools, NDK); Gradle 9.5 + Kotlin 2.3.20; **AGP 8.5.2 + plain kotlin.android works on Gradle 9.5; Kotlin MPP does NOT** (needs Gradle 8.x). pandoc/mmdc present; no LaTeX/drawio/plantuml/graphviz/docker. Live PostgreSQL server available. gh+g lab authed. API rate limits are transient — keep waves ≤ 6 agents, don't overlap workflows.